

PAVAN KUMAR P H V

Senior Applied AI Engineer | Forward Deployed AI Engineering

Bengaluru, India | Open to global remote roles from India

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PROFILE

Build AI systems that help finance and operations teams get reliable answers from business data. Combine **financial analytics and enterprise integration** with AI agents, production software and real-time computer vision. Work directly with stakeholders to define the business problem, implement the solution and validate results against source systems. Experience spans cloud applications, retail vision and industrial inspection.

BUSINESS & ENGINEERING STRENGTHS

Finance + AI agents: Translate margin, cost-to-serve, inventory, receivables/payables and cash-conversion metrics into analytical tools that agents can use to answer business questions.

Enterprise data + trustworthy answers: Work across SAP analytics, SQL data and document retrieval; validate calculations and AI outputs using source reconciliation, regression tests and LLM observability.

Computer vision + production ML: Build systems for retail recognition and industrial inspection; optimize GPU inference and deploy ML-serving APIs, including vision responses under 200 ms.

Customer delivery + cloud engineering: Bridge business stakeholders and engineering teams through requirements, architecture, implementation and debugging; cloud expertise across AWS, Azure and GCP.

PROFESSIONAL EXPERIENCE

Cervello India (a Kearney company)

Feb 2026 – Present

Senior Integration Consultant – Analytics & Information Management | Bengaluru

Enterprise AI for finance and supply-chain analytics

- Built **LangGraph agents for finance and operations questions**: classified intent, selected analytical tools and assembled calculated KPIs, charts and explanations from enterprise data.
- Translated stakeholder definitions of **margin, cost-to-serve and working capital** into Python services over SAP Datasphere/OData; resolved reporting-grain and fiscal-period questions before validating outputs.
- Checked that **AI-delivered financial figures matched source data**, tracing discrepancies from extraction and calculations through tools to the UI; used automated reconciliation, regression tests and Langfuse traces.

AI for process documentation and reusable business knowledge

- Helped turn business-process recordings into reusable knowledge: step-by-step guides, workflow diagrams, test cases and searchable documentation.
- Engineered asynchronous job orchestration, repository APIs, semantic chunking, embeddings and pgvector ingestion; implemented **hybrid semantic + SQL retrieval**, evidence synthesis and persistent task-scoped chat history.
- Hardened retrieval and execution through SQL parameterization, status normalization, failure handling and runtime decoupling; contributed across architecture, implementation and production debugging.

Technical leadership: Mentor two engineers and conduct first-round AI engineering interviews.

PROFESSIONAL EXPERIENCE, CONTINUED

SOLUM

Aug 2024 – Jan 2026

AI/CV & ML Systems Engineer | Bengaluru

- Designed multi-camera retail shelf-vision pipelines using YOLOv7/v9, TensorRT and LightGlue, with **sub-200 ms response times at 15–20 FPS** through GPU-optimized execution.
- Built RFID ML-serving APIs with Flask/Gunicorn, Azure Event Hubs and MongoDB; enabled hot-reload model updates, production observability and containerized deployment.
- Developed LLM log intelligence using fuzzy clustering, ReAct-based LangChain agents and locally hosted LLaMA models, supporting **1,000+ review cycles** and accelerating investigation workflows.
- Integrated model-serving and vision components into production delivery workflows, including CI/CD and operational monitoring.

Ignitarium

Aug 2022 – Jul 2024

AI Engineer | Bengaluru

- Refined the FastestDet architecture, improving detection accuracy by approximately **35%** while targeting efficient real-time deployment on resource-constrained Renesas microcontrollers.
- Built and optimized object-detection and segmentation models for intelligent embedded vision sensors; applied transfer learning and worked with hardware and software teams on deployment integration.
- Developed RGB-D reconstruction and binocular-vision pipelines for industrial defect detection, alongside domain-specific LLM question-answering and vector-retrieval applications.

Ignitarium

Jun 2021 – Jul 2022

AI Intern | Bengaluru

Developed and evaluated computer-vision and machine-learning prototypes, including model optimization and deployment-oriented experiments, before transitioning into a full-time AI engineering role.

TECHNICAL SKILLS

AI & retrieval: LangGraph, LangChain, ReAct, RAG, context engineering, embeddings, pgvector, Langfuse, evaluation and regression testing.

Software & data: Python, SQL, Go, C++, FastAPI, Flask/Gunicorn, PostgreSQL, Redis, MongoDB, REST APIs, async jobs, Git, PyTest.

Cloud & deployment: AWS (SageMaker, S3, Lambda, EC2), Azure (AKS, Event Hubs, Blob, Entra), GCP/Vertex AI, Docker, Kubernetes, CI/CD.

Vision & edge: TensorRT, ONNX, OpenCV, Open3D, YOLOv7/v9, LightGlue, object detection, segmentation, 3D reconstruction and embedded inference.

EDUCATION

M.Tech, Data Science

2022

Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Coimbatore

B.Tech, Mechanical Engineering

2019

Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Coimbatore

PROFESSIONAL CERTIFICATIONS

Google Cloud Certified – Professional Cloud Architect

2026

[View credential](#)

Anthropic Claude Certified Architect – Professional

2026

[View credential](#)